

Computer Networks — Lecture 01

Computer Networks and the Internet: Network Edge and Packet Switching (1.1 - 1.3.1)

المحاضرة 1 — مقدمة الإنترنت وحافة الشبكة وتبديل الحزم

Part 1: Multiple Choice Questions

1. In Internet jargon, what are the PCs, smartphones, TVs, and other devices connected to the Internet collectively called?

في مصطلحات الإنترنت، ماذا تسمى الأجهزة المتصلة بالإنترنت (حواسيب، هواتف، تلفزيونات...)?

- A) Edge routers
- B) Hosts or end systems
- C) Packet switches
- D) Access points

✓ **الإجابة الصحيحة:** B) Hosts or end systems

► **ليه صح:** كل الأجهزة الطرفية تسمى hosts أو end systems. الـ packet switches هي أجهزة التحويل في الشبكة نفسها وليست الأجهزة الطرفية، والـ edge router هو أول راوتر بعد جهاز المستخدم وليس اسمًا لكل الأجهزة.

● Ref: Lecture 01 — Section 1.1.1

2. Which pair represents the two most prominent types of packet switches in today's Internet?

ما هما أبرز نوعين من packet switches في الإنترنت اليوم؟

- A) DSLAMs and CMTSSs
- B) Modems and splitters
- C) Hosts and servers
- D) Routers and link-layer switches

✓ **الإجابة الصحيحة:** D) Routers and link-layer switches

► **ليه صح:** المحاضرة تذكر أن أبرز نوعين هما routers و link-layer switches. الـ DSLAM والـ CMTS أجهزة في شبكات الوصول لتحويل الإشارات، والـ hosts والـ servers أجهزة طرفية وليست switches.

● Ref: Lecture 01 — Section 1.1.1

3. Where are link-layer switches and routers typically used?

أين يُستخدم كل من link-layer switches وrouters عادةً؟

- A) Both only in access networks
- B) Link-layer switches in access networks; routers in the network core
- C) Both only inside data centers
- D) Routers in access networks; link-layer switches in the network core

✓ **الإجابة الصحيحة:** B) Link-layer switches in access networks; routers in the network core

► **ليه صح:** ال link-layer switches تُستخدم عادةً في شبكات الوصول (access networks) بينما ال routers تُستخدم في قلب الشبكة (network core). الاختيار المعكوس هو الأقرب للصح لكنه عكس ما في المحاضرة.

● Ref: Lecture 01 — Section 1.1.1

4. What is the sequence of communication links and packet switches traversed by a packet from the sending end system to the receiving end system called?

ماذا يسمى تسلسل الروابط وال packet switches الذي تمر به الحزمة من المرسل إلى المستقبل؟

- A) A socket interface
- B) An access network
- C) A protocol
- D) A route (or path) through the network

✓ **الإجابة الصحيحة:** D) A route (or path) through the network

► **ليه صح:** هذا التسلسل يسمى route أو path. ال protocol يحدد صيغة وترتيب الرسائل وليس المسار، وال access network هي الشبكة التي توصل الجهاز بأول راوتر فقط.

● Ref: Lecture 01 — Section 1.1.1

5. Which organization develops Internet standards, whose documents are called RFCs?

أي جهة تضع معايير الإنترنت التي تسمى وثائقها RFCs؟

- A) The Transmission Control Protocol
- B) The IEEE 802 LAN Standards Committee
- C) An access ISP
- D) The Internet Engineering Task Force (IETF)

✓ **الإجابة الصحيحة:** D) The Internet Engineering Task Force (IETF)

► **ليه صح:** ال IETF هي التي تضع معايير الإنترنت وتسمى وثائقها RFCs. لجنة IEEE 802 تحدد معايير Ethernet وWiFi فقط (معايير الروابط) وليس معايير الإنترنت.

● Ref: Lecture 01 — Section 1.1.1

6. Which statement about Internet applications is correct?

أي عبارة صحيحة عن تطبيقات الإنترنت؟

- A) They run on end systems, not on the packet switches in the network core
- B) They run inside routers in the network core
- C) They run only on servers inside data centers
- D) They run inside link-layer switches

✓ **الإجابة الصحيحة:** A) They run on end systems, not on the packet switches in the network core

► **ليه صح:** تطبيقات الإنترنت تعمل على الأجهزة الطرفية (end systems) وليس على الـ packet switches في قلب الشبكة. وهي لا تقتصر على الخوادم، فالعملاء (clients) يشغلونها أيضًا.

● Ref: Lecture 01 — Section 1.1.2

7. What is the Internet socket interface?

ما هي واجهة الـ socket في الإنترنت؟

- A) A set of rules that a sending program must follow so the Internet can deliver its data to a specific destination program
- B) A protocol that automatically sets the forwarding tables
- C) A physical port on a router that joins two links
- D) A header field that stores the destination IP address

✓ **الإجابة الصحيحة:** A) A set of rules that a sending program must follow so the Internet can deliver its data to a specific destination program

► **ليه صح:** الـ socket interface مجموعة قواعد يتبعها البرنامج المرسل ليسلم الإنترنت بياناته إلى برنامج وجهة محدد. الاختيار الأخير هو وصف لـ routing protocols وليس للـ socket.

● Ref: Lecture 01 — Section 1.1.2

8. According to the lecture, what does a protocol define?

حسب المحاضرة، ماذا يحدد الـ protocol؟

- A) Only the physical medium used to carry bits
- B) Only the format of the messages exchanged
- C) Only the transmission rate of a link
- D) The format and order of messages exchanged, and the actions taken on transmission or receipt of a message or other event

✓ **الإجابة الصحيحة:** D) The format and order of messages exchanged, and the actions taken on transmission or receipt of a message or other event

► **ليه صح:** الـ protocol يحدد صيغة وترتيب الرسائل المتبادلة والإجراءات المتخذة عند الإرسال أو الاستقبال أو حدوث حدث ما. الاختيار الأول ناقص لأنه يهمل الترتيب والإجراءات.

● Ref: Lecture 01 — Section 1.1.3

9. In the Web request example, what does your computer send right after receiving the connection reply message from the Web server?

في مثال طلب صفحة ويب، ماذا يرسل جهازك بعد استلام رسالة connection reply؟

- A) A GET message containing the name of the Web page it wants
- B) Another connection request message
- C) The Web page file itself
- D) The IP address of the nearest router

✓ **الإجابة الصحيحة:** A) A GET message containing the name of the Web page it wants

► **ليه صح:** بعد رد الاتصال يرسل الجهاز رسالة GET فيها اسم الصفحة المطلوبة، ثم يرد الخادم بالصفحة. إرسال الصفحة هو دور الخادم وليس العميل.

● Ref: Lecture 01 — Section 1.1.3

10. What is an access network?

ما هي شبكة الوصول (access network)؟

- A) The set of protocols that route packets between hosts
- B) The mesh of packet switches and links at the center of the Internet
- C) The network that interconnects different ISPs with each other
- D) The network that physically connects an end system to the first (edge) router on a path to any other end system

✓ **الإجابة الصحيحة:** D) The network that physically connects an end system to the first (edge) router on a path to any other end system

► **ليه صح:** شبكة الوصول هي التي توصل الجهاز الطرفي فعليًا بأول راوتر (edge router) في المسار. الاختيار الأول يصف الـ network core وليس الـ access network.

● Ref: Lecture 01 — Section 1.2.1

11. With DSL Internet access, where is the DSLAM located?

في خدمة DSL، أين يوجد جهاز DSLAM؟

- A) At the cable head end
- B) In the telco's local central office (CO)
- C) Inside the customer's home
- D) At a neighborhood fiber junction

✓ **الإجابة الصحيحة:** B) In the telco's local central office (CO)

► **ليه صح:** الـ DSLAM يوجد في المقسم المحلي لشركة الاتصالات (central office). الـ cable head end والـ fiber junction ينتميان لشبكة الكابل (HFC) وليس DSL.

● Ref: Lecture 01 — Section 1.2.1

12. How does a residential DSL telephone line carry both data and traditional telephone signals at the same time?

كيف يحمل خط DSL السكني البيانات وإشارات الهاتف التقليدية في نفس الوقت؟

- A) They alternate in fixed time slots
- B) The phone signal is first converted into packets
- C) They are encoded at different frequencies
- D) They travel on two separate optical fibers

✓ **الإجابة الصحيحة:** C) They are encoded at different frequencies

► **ليه صح:** تُرمّز البيانات وإشارة الهاتف على ترددات مختلفة، ويفصلهما الـ splitter عند العميل والـ DSLAM عند الشركة. تقسيم الوقت (time slots) ليس هو الأسلوب المذكور هنا.

● Ref: Lecture 01 — Section 1.2.1

13. Why is cable Internet access often called hybrid fiber coax (HFC)?

لماذا تسمى خدمة الإنترنت عبر الكابل (HFC) hybrid fiber coax؟

- A) Fiber connects the cable head end to neighborhood junctions, then coaxial cable reaches individual homes
- B) Satellite links and coaxial cable are combined
- C) Fiber runs to every house, with coaxial cable only inside the house
- D) Twisted-pair copper reaches the junctions and fiber reaches the homes

✓ **الإجابة الصحيحة:** A) Fiber connects the cable head end to neighborhood junctions, then coaxial cable reaches individual homes

► **ليه صح:** لأنها تجمع الألياف الضوئية من الـ head end إلى تقاطعات الأحياء ثم كابل متحد المحور (coaxial) إلى المنازل. الألياف التي تصل لكل منزل تسمى FTTH وليست HFC.

● Ref: Lecture 01 — Section 1.2.1

14. At the cable head end, which device plays a role similar to the DSLAM in a DSL network?

في الـ cable head end، أي جهاز يؤدي وظيفة مشابهة للـ DSLAM في شبكة DSL؟

- A) The splitter
- B) The cable modem termination system (CMTS)
- C) The cable modem
- D) The Ethernet switch

✓ **الإجابة الصحيحة:** B) The cable modem termination system (CMTS)

► **ليه صح:** الـ CMTS يحوّل الإشارة التناظرية القادمة من الـ cable modems في المنازل إلى رقمية، تمامًا كما يفعل الـ DSLAM. الـ cable modem جهاز عند العميل وليس عند الـ head end.

● Ref: Lecture 01 — Section 1.2.1

15. Why might a user's actual download rate be much lower than the aggregate cable downstream rate?

لماذا قد تكون سرعة التحميل الفعلية للمستخدم أقل بكثير من السرعة الكلية لقناة الكابل؟

- A) Each home is given a fixed dedicated slice of the downstream rate
- B) The upstream channel is always faster and throttles the downstream channel
- C) Cable is a shared broadcast medium, so users downloading simultaneously share the downstream channel
- D) Coaxial cable cannot carry digital data

✓ **الإجابة الصحيحة:** C) Cable is a shared broadcast medium, so users downloading simultaneously share the downstream channel

► **ليه صح:** الكابل وسط مشترك (shared broadcast medium) فعند تحميل عدة مستخدمين معًا يتقاسمون القناة. القناة الهابطة عادةً أسرع من الصاعدة وليس العكس.

● Ref: Lecture 01 — Section 1.2.1

16. Which technology is by far the most prevalent access technology in corporate, university, and home LANs?

أي تقنية هي الأكثر انتشارًا بفارق كبير في شبكات LAN للشركات والجامعات والمنازل؟

- A) WiFi (IEEE 802.11)
- B) DSL
- C) Fiber to the home (FTTH)
- D) Ethernet

✓ **الإجابة الصحيحة:** D) Ethernet

► **ليه صح:** ال Ethernet هو الأكثر انتشارًا، ويستخدم أسلاك twisted-pair للوصل إلى Ethernet switch. ال WiFi منتشر لكنه ليس الأكثر، وDSL وFTTH تقنيات وصول سكني.

● Ref: Lecture 01 — Section 1.2.1

17. Unlike WiFi, a cellular wide-area wireless user need only be within roughly how far of the base station?

على عكس WiFi، المستخدم في الشبكات الخلوية يجب أن يكون ضمن مسافة تقريبية من المحطة الأساسية تبلغ؟

- A) A few tens of kilometers
- B) One or two meters
- C) A few hundred kilometers
- D) A few tens of meters

✓ **الإجابة الصحيحة:** A) A few tens of kilometers

► **ليه صح:** المدى في الشبكات الخلوية عشرات الكيلومترات، بينما WiFi مداه عشرات الأمتار. المتر أو المترين هو مدى الأجهزة الشخصية القريبة مثل سماعات لاسلكية.

● Ref: Lecture 01 — Section 1.2.1

18. Which of the following is an unguided physical medium?

أي مما يلي وسط فيزيائي غير موجّه (unguided)؟

- A) Twisted-pair copper wire
- B) A digital satellite channel
- C) Coaxial cable
- D) Fiber-optic cable

✓ **الإجابة الصحيحة:** B) A digital satellite channel

► **ليه صح:** في الوسائط غير الموجهة تنتشر الموجات في الجو أو الفضاء مثل القناة الفضائية للأقمار الصناعية. الباقي كلها وسائط موجهة (guided) تمر فيها الموجات داخل وسط صلب.

● Ref: Lecture 01 — Section 1.2.2

19. Which is the least expensive and most commonly used guided transmission medium?

أي وسط نقل موجّه هو الأرخص والأكثر استخدامًا؟

- A) Multimode fiber with optical switches
- B) Coaxial cable
- C) Twisted-pair copper wire
- D) Fiber-optic cable

✓ **الإجابة الصحيحة:** C) Twisted-pair copper wire

► **ليه صح:** ال twisted-pair copper wire هو الأرخص والأكثر شيوعًا، ويُستخدم في الهاتف وال LANs. ال coaxial أغلى منه ويستخدم أساسًا في التلفزيون الكابلي.

● Ref: Lecture 01 — Section 1.2.2

20. What has hindered the deployment of fiber optics for short-haul transport such as LANs or residential access?

ما الذي أعاق استخدام الألياف الضوئية في المسافات القصيرة كال LANs والوصول السكني؟

- A) Its susceptibility to electromagnetic interference
- B) It cannot support rates above 100 Mbps
- C) The high cost of optical devices such as transmitters, receivers, and switches
- D) Its very high signal attenuation over short distances

✓ **الإجابة الصحيحة:** C) The high cost of optical devices such as transmitters, receivers, and switches

► **ليه صح:** السبب هو التكلفة العالية للأجهزة الضوئية (مرسلات ومستقبلات ومحولات). الألياف في الحقيقة محصنة ضد التداخل الكهرومغناطيسي ولها توهين منخفض جدًا، فالخيارات الأخرى تناقض خصائصها.

● Ref: Lecture 01 — Section 1.2.2

21. A geostationary satellite is placed at about what altitude, and what substantial propagation delay does this introduce?

على أي ارتفاع تقريبي يوضع القمر الجيوستاشنري وما تأخير الانتشار الكبير الناتج عن ذلك؟

- A) About 36,000 km; roughly 280 milliseconds
- B) About 3,600 km; roughly 280 milliseconds
- C) About 360 km; roughly 2.8 milliseconds
- D) About 36,000 km; roughly 28 milliseconds

✓ **الإجابة الصحيحة:** A) About 36,000 km; roughly 280 milliseconds

► **ليه صح:** القمر الثابت بالنسبة للأرض يوضع على ارتفاع 36,000 كم وينتج تأخير انتشار حوالي 280 ملي ثانية. الأرقام الأخرى تغيّر الارتفاع أو التأخير بمقدار عشر مرات.

● Ref: Lecture 01 — Section 1.2.2

22. What does store-and-forward transmission mean for a packet switch?

ماذا يعني الإرسال بأسلوب store-and-forward في ال packet switch؟

- A) The switch stores packets permanently for later retrieval
- B) The switch forwards each bit immediately as it arrives
- C) The switch must receive the entire packet before it begins transmitting the first bit onto the outbound link
- D) The switch forwards only the packet header first, then the data

✓ **الإجابة الصحيحة:** C) The switch must receive the entire packet before it begins transmitting the first bit onto the outbound link

► **ليه صح:** يجب أن يستقبل ال switch الحزمة كاملة قبل أن يبدأ إرسال أول bit منها على الرابط الخارج. إعادة الإرسال الفورية لكل bit تعاكس هذا الأسلوب.

● Ref: Lecture 01 — Section 1.3.1

23. In the figure with Hosts A and B sending to Host E over 100 Mbps links into a router with a 15 Mbps outgoing link, what happens if the packet arrival rate exceeds 15 Mbps for a short interval?

في الشكل الذي يرسل فيه A و B إلى E عبر روابط 100 Mbps إلى راوتر رابطته الخارج 15 Mbps، ماذا يحدث إذا تجاوز معدل وصول الحزم 15 Mbps لفترة قصيرة؟

- A) Congestion occurs and packets queue in the output buffer of the 15 Mbps link
- B) The packets bypass the router
- C) The packets are immediately forwarded on the 100 Mbps links
- D) The router increases its outgoing link rate to match

✓ **الإجابة الصحيحة:** A) Congestion occurs and packets queue in the output buffer of the 15 Mbps link

► **ليه صح:** يحدث ازدحام (congestion) وتنتظر الحزم في ال output buffer قبل الإرسال على رابط ال 15 Mbps. معدل الرابط ثابت ولا يتغير تلقائيًا.

● Ref: Lecture 01 — Section 1.3.1

24. When does packet loss occur at a router?

متى يحدث فقد الحزم (packet loss) في الراوتر؟

- A) When the propagation speed of the link is too low
 - B) When an arriving packet finds the output buffer completely full, so a packet must be dropped
 - C) When the forwarding table contains too many entries
 - D) When the link transmission rate is higher than the arrival rate
- ✓ **الإجابة الصحيحة:** B) When an arriving packet finds the output buffer completely full, so a packet must be dropped

► **ليه صح:** يحدث الفقد عندما تصل حزمة وال buffer ممتلئ تمامًا، فتُسقط الحزمة الواصلة أو إحدى الحزم المنتظرة. لو كان معدل الإرسال أعلى من معدل الوصول فلن تتكون طوابير ممتلئة أصلاً.

● Ref: Lecture 01 — Section 1.3.1

25. What does a router's forwarding table do?

ماذا يفعل جدول التوجيه (forwarding table) في الراوتر؟

- A) It maps application names to port numbers
 - B) It stores copies of packets until the destination acknowledges them
 - C) It maps destination addresses (or portions of them) to the router's outbound links
 - D) It lists the ISPs that the router pays for connectivity
- ✓ **الإجابة الصحيحة:** C) It maps destination addresses (or portions of them) to the router's outbound links

► **ليه صح:** ال forwarding table يربط عناوين الوجهة (أو جزءًا منها) بالروابط الخارجة من الراوتر، ويبحث فيه الراوتر بعنوان ال IP للوجهة. ال routing protocols هي التي تملأ الجداول تلقائيًا وليس الجدول نفسه.

● Ref: Lecture 01 — Section 1.3.1

Part 2: Essay and Calculation Questions

1. Describe the Internet from the "nuts-and-bolts" point of view.

صف الإنترنت من منظور المكونات الأساسية (nuts-and-bolts).

Model answer:

- The Internet is a computer network that interconnects billions of computing devices (hosts or end systems): PCs, servers, smartphones, tablets, TVs, cars, thermostats, etc.
- End systems are connected by communication links (with different transmission rates in bits/second) and packet switches.
- The sender segments data and adds header bytes to form packets; packets travel through the network and are reassembled at the destination.
- Packet switches (mainly routers and link-layer switches) forward packets; the links and switches traversed form a route or path.
- End systems access the Internet through ISPs, which are interconnected themselves, and all pieces run protocols such as TCP and IP (TCP/IP), standardized by the IETF (RFCs) and the IEEE 802 committee (Ethernet, WiFi).

● Ref: Lecture 01 — Section 1.1.1

2. Describe the Internet as an infrastructure that provides services to applications. What is a distributed application and what is the socket interface?

صف الإنترنت كبنية تحتية تقدم خدمات للتطبيقات. ما هو التطبيق الموزع (distributed application) وما هي واجهة الـ socket؟

Model answer:

- Applications such as e-mail, Web, messaging, streaming, video conferencing and games use the Internet as a platform.
- They are distributed applications because they involve multiple end systems exchanging data with each other.
- Applications run on end systems, not in the packet switches of the network core.
- Programs on different end systems must send data to each other, so end systems provide a socket interface: a set of rules the sending program must follow so the Internet delivers data to a specific destination program.
- The Internet provides multiple services, and the developer must choose one of them for the application.

● Ref: Lecture 01 — Section 1.1.2

3. What is a protocol? Explain with a human analogy and a Web request example.

ما هو البروتوكول؟ اشرحه بمثال بشري ومثال طلب صفحة ويب.

Model answer:

- A protocol defines the format and the order of messages exchanged between two or more communicating entities, as well as the actions taken on transmission/receipt of a message or other event.
- Human analogy: asking for the time. First say "Hi"; a returned "Hi" means you can proceed and ask for the time. A different reply, or no reply, means you do not ask (or give up).
- Web example: the computer sends a connection request, waits for a connection reply, then sends a GET message with the page name, and finally the server returns the page.
- All activity in the Internet involving two or more remote communicating entities is governed by a protocol.

● Ref: Lecture 01 — Section 1.1.3

4. Define hosts, end systems, clients, and servers, and explain what the network edge is.

عرّف الـ hosts والـ end systems والـ clients والـ servers، واذّرح ما هي حافة الشبكة (network edge).

Model answer:

- Computers and other devices connected to the Internet are called end systems because they sit at the edge of the Internet.
- They are also called hosts because they host (run) application programs such as a Web browser, Web server, or e-mail client/server. In this book host = end system.
- Hosts are divided into clients (desktops, laptops, smartphones) and servers (more powerful machines that store and distribute Web pages, stream video, relay e-mail).
- Most servers providing search results, e-mail, Web pages, videos and mobile app content reside in large data centers.

● Ref: Lecture 01 — Section 1.2

5. Compare DSL and cable Internet access.

قارن بين الإنترنت عبر DSL والإنترنت عبر الكابل.

Model answer:

- DSL: uses the existing telephone line from the same telco that is the ISP. The DSL modem exchanges data with a DSLAM in the telco's central office; data and phone signals share the line at different frequencies and a splitter separates them. Designed for short distances (generally within 5 to 10 miles of the CO).
- Cable: uses the cable TV infrastructure. Fiber connects the head end to neighborhood junctions (500 to 5,000 homes each), then coaxial cable reaches homes; hence hybrid fiber coax (HFC). It needs cable modems and a CMTS at the head end (similar role to the DSLAM).
- Both are typically asymmetric: the downstream channel gets a higher rate than the upstream.
- Cable is a shared broadcast medium, so simultaneous heavy downloads reduce each user's rate; with few active users each may get the full downstream rate.

● Ref: Lecture 01 — Section 1.2.1

6. Explain access in enterprises and homes (Ethernet and WiFi) and wide-area wireless access.

اشرح الوصول في الشركات والمنازل (WiFi و Ethernet) والوصول اللاسلكي واسع النطاق.

Model answer:

- A local area network (LAN) connects an end system to the edge router. Ethernet is by far the most prevalent: users connect with twisted-pair copper wire to an Ethernet switch, which connects to the larger Internet.
- In a wireless LAN, users send/receive packets to/from an access point connected to the enterprise network; users must be within a few tens of meters. WiFi is based on IEEE 802.11.
- Many homes combine broadband residential access (cable modem or DSL) with wireless LAN.
- Wide-area wireless access uses the cellular infrastructure: devices send packets through a base station operated by the cellular provider, within a few tens of kilometers. 4G gives real-world speeds up to 60 Mbps, and 5G is being deployed.

● Ref: Lecture 01 — Section 1.2.1

7. Distinguish between guided and unguided media, and describe twisted pair, coaxial cable, and fiber optics.

ميّز بين الوسائط الموجهة وغير الموجهة، وصف ال twisted pair وال coaxial cable والألياف الضوئية.

Model answer:

- Guided media: waves are guided along a solid medium (fiber-optic cable, twisted-pair copper wire, coaxial cable). Unguided media: waves propagate in the atmosphere or outer space (wireless LAN, digital satellite channel).
- Twisted pair: the least expensive and most common guided medium; two insulated copper wires in a spiral; UTP is used for LANs (10 Mbps to 10 Gbps; category 6a reaches 10 Gbps up to a hundred meters).
- Coaxial cable: two concentric copper conductors; high rates; common in cable TV and cable Internet; can be a guided shared medium.
- Fiber optics: thin flexible medium conducting light pulses; tens or hundreds of Gbps; immune to electromagnetic interference, very low attenuation up to 100 km, hard to tap. Preferred for long-haul and the Internet backbone, but optical device costs hinder short-haul use.

● Ref: Lecture 01 — Section 1.2.2

8. Describe terrestrial radio channels and satellite radio channels, including the two types of communication satellites.

صف قنوات الراديو الأرضية وقنوات الأقمار الصناعية مع نوعي الأقمار الصناعية المستخدمة في الاتصالات.

Model answer:

- Terrestrial radio channels carry signals in the electromagnetic spectrum, need no physical wire, can penetrate walls, support mobile users and can carry signals long distances.
- They are classified by range: very short (one or two meters, e.g., wireless headsets/keyboards), local area (ten to a few hundred meters, wireless LANs), and wide area (tens of kilometers, cellular).
- A communication satellite links two or more ground stations: it receives on one frequency band, regenerates the signal with a repeater, and transmits on another frequency.
- Geostationary satellites stay above the same spot at 36,000 km, causing about 280 ms propagation delay; they are used where DSL or cable is unavailable. LEO satellites are much closer to Earth, orbit like the Moon, and may communicate with each other and with ground stations.

● Ref: Lecture 01 — Section 1.2.2

9. Explain store-and-forward transmission and derive the delay for one packet over two links and for three packets.

اشرح أسلوب الإرسال store-and-forward واستنتج التأخير لحزمة واحدة عبر رابطتين وثلاث حزم.

Model answer:

- Store-and-forward: the packet switch must receive the entire packet before it can begin to transmit its first bit onto the outbound link; it buffers the bits first.
- One packet of L bits, two links of rate R , one router: the source finishes at time L/R , the router starts forwarding at L/R and finishes at $2L/R$. Total delay = $2L/R$.
- Three packets: at L/R the router forwards packet 1 while the source starts packet 2; at $2L/R$ the destination has packet 1 and the router has packet 2; at $3L/R$ the destination has two packets and the router has packet 3; at $4L/R$ the destination has all three.
- General case: N links each of rate R ($N-1$ routers) gives $d(\text{end-to-end}) = N L/R$.

● Ref: Lecture 01 — Section 1.3.1

10. Explain output buffers, queuing delay, packet loss, and how forwarding tables and routing protocols work in the Internet.

اشرح الـ output buffers وتأخير الطابور وفقد الحزم وكيف تعمل جداول التوجيه وبروتوكولات التوجيه في الإنترنت.

Model answer:

- Each packet switch has an output buffer (output queue) for each attached link, storing packets about to be sent onto that link.
- If an arriving packet finds the link busy, it waits in the output buffer, so packets suffer queuing delays in addition to store-and-forward delays. These delays are variable and depend on congestion.
- The buffer is finite: if it is completely full, packet loss occurs and either the arriving packet or an already-queued packet is dropped.
- Forwarding: each end system has an IP address; the source puts the destination IP address in the packet header. Each router has a forwarding table mapping destination addresses to outbound links; it examines the address, searches the table and directs the packet to the appropriate outbound link.
- The Internet has special routing protocols that automatically set the forwarding tables.

● Ref: Lecture 01 — Section 1.3.1

11. A source sends one packet of 1,500 bytes to a destination through one router. Both links have rate $R = 10$ Mbps. Assuming store-and-forward and ignoring processing, queuing and propagation delays, what is the total delay from the start of transmission until the whole packet is received?

يرسل مصدر حزمة واحدة حجمها 1,500 بايت إلى وجهة عبر راوتر واحد، ومعدل كل رابط 10 Mbps. بافتراض store-and-forward وإهمال باقي التأخيرات، ما التأخير الكلي؟

Model answer:

- Given: $L = 1,500 \text{ bytes} \times 8 = 12,000 \text{ bits}$; $R = 10 \text{ Mbps} = 10,000,000 \text{ bits/s}$; $N = 2$ links.
- Formula: $d(\text{end-to-end}) = N L/R$.
- $L/R = 12,000 / 10,000,000 = 0.0012 \text{ s} = 1.2 \text{ ms}$.
- $d = 2 \times 1.2 \text{ ms} = 2.4 \text{ ms}$.

● Ref: Lecture 01 — Section 1.3.1

12. A packet of 12,000 bits crosses two links of equal rate R with one store-and-forward router in between. The measured total delay is 6 ms (ignoring other delays). Find R .

حزمة حجمها 12,000 bit تعبر رابطتين لهما نفس المعدل R مع راوتر store-and-forward بينهما، والتأخير الكلي المقاس 6 ms (بإهمال باقي التأخيرات). أوجد R .

Model answer:

- Given: $L = 12,000 \text{ bits}$; $N = 2$; $d = 6 \text{ ms} = 0.006 \text{ s}$.
- Formula: $d = N L/R$, so $R = N L / d$.
- $R = 2 \times 12,000 / 0.006 = 24,000 / 0.006 = 4,000,000 \text{ bits/s}$.
- $R = 4 \text{ Mbps}$.

● Ref: Lecture 01 — Section 1.3.1

13. A source sends three packets, each of $L = 10,000$ bits, to a destination through one router. Both links have $R = 1$ Mbps. Using store-and-forward, when has the destination received all three packets?

يرسل مصدر ثلاث حزم، كل منها $L = 10,000 \text{ bit}$ ، إلى وجهة عبر راوتر واحد. معدل الرابطتين $R = 1 \text{ Mbps}$. متى تستلم الوجهة الحزم الثلاث كلها؟

Model answer:

- Given: $L = 10,000 \text{ bits}$; $R = 1,000,000 \text{ bits/s}$; 3 packets; 2 links.
- $L/R = 10,000 / 1,000,000 = 0.01 \text{ s} = 10 \text{ ms}$.
- From the lecture example: the destination has received all three packets at time $4L/R$.
- Total = $4 \times 10 \text{ ms} = 40 \text{ ms}$.

● Ref: Lecture 01 — Section 1.3.1

14. Host A sends a packet of 1,500 bytes over a 100 Mbps Ethernet link to a router, which forwards it onto a 15 Mbps link. Ignoring all other delays, how long does it take for the packet to be completely received at the next node after the router (store-and-forward)?

يرسل Host A حزمة 1,500 بايت عبر رابط Ethernet سرعته 100 Mbps إلى راوتر يمررها على رابط 15 Mbps. بإهمال باقي التأخيرات، كم يستغرق وصول الحزمة كاملة إلى العقدة التالية بعد الراوتر؟

Model answer:

- Given: $L = 1,500 \times 8 = 12,000$ bits; $R_1 = 100$ Mbps; $R_2 = 15$ Mbps.
- Formula: transmission time on a link = L/R ; the router must receive the whole packet first (store-and-forward), so the two times add.
- First link: $12,000 / 100,000,000 = 0.00012$ s = 0.12 ms.
- Second link: $12,000 / 15,000,000 = 0.0008$ s = 0.8 ms.
- Total = $0.12 + 0.8 = 0.92$ ms.

● Ref: Lecture 01 — Section 1.3.1

15. A path from source to destination has $N = 5$ links, each of rate $R = 3$ Mbps. One packet of $L = 6,000$ bits is sent. How many routers are on the path, and what is the end-to-end delay (store-and-forward, other delays ignored)?

مسار من المصدر للوجهة فيه $N = 5$ روابط، معدل كل منها $R = 3$ Mbps. تُرسل حزمة واحدة $L = 6,000$ bit. كم راوتر على المسار وما التأخير من طرف إلى طرف؟

Model answer:

- Given: $N = 5$ links; $R = 3,000,000$ bits/s; $L = 6,000$ bits.
- Number of routers = $N - 1 = 4$.
- $L/R = 6,000 / 3,000,000 = 0.002$ s = 2 ms.
- Formula: $d(\text{end-to-end}) = N L/R = 5 \times 2$ ms = 10 ms.

● Ref: Lecture 01 — Section 1.3.1